

## Research Paper

## Enhancement of syngas through integrating carbon dioxide in the catalytic pyrolysis of plantation waste

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## ARTICLE INFO

## Keywords:

Circular economy  
Waste valorisation  
Waste-to-energy  
Rubber tree waste  
Pyrolysis  
CO<sub>2</sub> utilization

## ABSTRACT

This study introduces the sustainable recovery of energy from plantation waste by converting rubber tree residue (RTR) into syngas through carbon dioxide (CO<sub>2</sub>)-assisted pyrolysis. The investigation centers on elucidating the specific impact of CO<sub>2</sub> on the pyrolysis of rubber tree residue, emphasizing its significant influence on enhancing syngas production. At temperatures  $\geq 500$  °C, CO<sub>2</sub> engages with volatile matters (VMs) released during the thermolysis of rubber tree residue, increasing carbon monoxide (CO) yields. To further optimize the process, catalytic pyrolysis setups incorporating additional heat (600 °C) and a nickel-based catalyst (Ni/Al<sub>2</sub>O<sub>3</sub>) were implemented. These experimental configurations substantially increased syngas production from 19.51 to 24.24 mmol g<sup>-1</sup>, particularly amplifying CO yields under CO<sub>2</sub> conditions 2.58-fold compared to nitrogen (N<sub>2</sub>) conditions. This enhancement is attributed to the partial oxidation of volatile matters facilitated by CO<sub>2</sub>. Additionally, the Ni/Al<sub>2</sub>O<sub>3</sub> catalyst played a pivotal role in expediting the gas-phase homogeneous reaction of CO<sub>2</sub> with volatile matters, leading to further improved syngas production. Indeed, the syngas yield in the catalytic pyrolysis in the presence of CO<sub>2</sub> was 24.24 mmol g<sup>-1</sup>, which increased by 1.24 times in reference to the result under the N<sub>2</sub> environment. The identified functional role of CO<sub>2</sub> presents an opportunity to enhance the sustainability of waste management by optimizing carbon utilization and generating value-added products.

## 1. Introduction

Global warming, resulting from the combustion of fossil fuels for primary energy production [1,2], has begun having tangible impacts on the world, notably manifested in climate change [3,4]. Recognising the urgency of this situation, reliance on fossil fuels needs rectifying and initiatives expedited to increase the share of renewable energy in primary energy generation [5]. Presently, governments in developed countries are providing financial support through legislation to foster the development of renewable energy resources, particularly solar and wind [6]. However, transitioning to renewable energy sources poses challenges, given the existing carbon-based infrastructure deeply ingrained in traditional industries [7]. An increasingly prominent alternative lies in harnessing biomass, a renewable resource abundant in

carbon and known for its effective mitigation of net greenhouse gas (GHG) emissions into the atmosphere [8]. Biomass comprises plant-based chemicals formed by the absorption of CO<sub>2</sub> via photosynthesis [9,10] and holds substantial potential as an energy source, accounting for a significant portion (13 %) of current energy consumption [11]. However, it is crucial to acknowledge the associated risks with biomass utilization, including environmental degradation arising from deforestation and competition for land, which could potentially compromise food security [12]. To address these concerns, a viable approach is to focus on exploiting residues (wastes) from large-scale centralized plantations as the primary source of biomass. These residues consist of lignocellulosic biomass (LB), typically comprising 20–45 wt% of cellulose, 20–40 wt% of hemicellulose, and 10–25 wt% of lignin [13].

One prevalent method for harnessing energy from LB residue (waste)

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involves direct combustion [14]. While LB is considered a sustainable primary resource, its combustion raises environmental concerns due to challenges in controlling the equivalence ratio between air and solid fuel [15,16]. The diffusion flame of air into LB leads to incomplete oxidation, resulting in the release of various toxic chemicals such as polycyclic aromatic hydrocarbons (PAHs), volatile organic compounds (VOCs), and unburned hydrocarbons (UHCs) [14]. To address these technical obstacles, processing LB into liquid and/or gaseous forms facilitates better control of the equivalence ratio [17], enhancing the combustion efficiency governed by premixed flame [18]. In the pursuit of these objectives, two representative technologies for biofuel production from LB have emerged, namely biochemical and thermochemical processes [15]. Biochemical processes, involving fermentation and anaerobic digestion, are well-defined approaches for producing bioethanol and biogas [19,20]. These processes ensure high yield and selectivity for desired products based on the efficiency of the conversion process [21,22]. However, due to the inherently heterogeneous nature of LB, rigorous pre-treatments (such as acid/alkaline hydrolysis) are imperative to obtain single-component reagents, such as monosaccharides [13,23]. The biochemical conversion of these reagents also requires a long reaction time and exhibits a slow reaction kinetics [24,25]. Moreover, microbial metabolisms inevitably generate biological byproducts, including sludge and  $\text{CO}_2$  [26], leading to a loss of available carbons from LB and a reduction in the productivity of target chemicals [27].

Thermochemical processes, such as gasification and pyrolysis, present promising solutions due to their adaptability to various raw materials. These processes have demonstrated the ability to reassign elemental carbon in LB into syngas, biocrude, and biochar while minimizing carbon loss [28]. A great deal of research on pyrolysis process has typically aimed at producing biocrude [29–31]. Thanks to its tailorable

carbon lengths and versatile compositional matrices, biocrude has functioned a promising building stock for chemical synthesis [32]. Despite their effectiveness, the complicated composition of biocrude poses challenges in its commercial implementation, necessitating further upgrading and separation processes [33]. These robust post-treatments could undermine an economic feasibility of utilizing biocrude [34]. Alternatively, converting biocrude into simpler gaseous forms, specifically syngas (a mixture of  $\text{H}_2$  and  $\text{CO}$ ), may be desirable option [35]. Since syngas is the end product of thermal decomposition, it is more accessible to obtain with high purity. Similar to biocrude, syngas also served as a direct fuel or an intermediate feedstock for the synthesis of versatile chemicals [36,37]. Although gasification is an effective route for transforming LB into syngas by reacting with oxidizing agents (air, oxygen, or water) [38], its widespread adoption is limited by the high-temperature requirement (typically activated at  $\geq 700^\circ\text{C}$ ) [39]. In this context, exploring a feasible technique for syngas production from LB at pyrolysis temperatures  $\leq 700^\circ\text{C}$  could be an effective alternative [38]. However, the lower temperatures reduce the degree of thermal cracking of VMs evolved from the thermolysis of LB, leading to increased biocrude yield. In addressing the challenges associated with biocrude, it is important to develop pyrolysis techniques aimed at its comprehensive transformation into syngas.

Based on these rationales, this study explores the thermal degradation of LB, with particular stress on enhancing syngas yield. Rubber tree residue (RTR) was chosen as a model plantation waste due to its substantial generation resulting from the continuous expansion of rubber plantations [40]. Given the commercial lifespan of rubber tree (approximately 30 years) [41], its annual generation is estimated at  $74.5 \times 10^9 \text{ kg}$  (as detailed in Table 4). Accordingly, RTR facilitates a reliable and consistent supply as the primary energy resource,

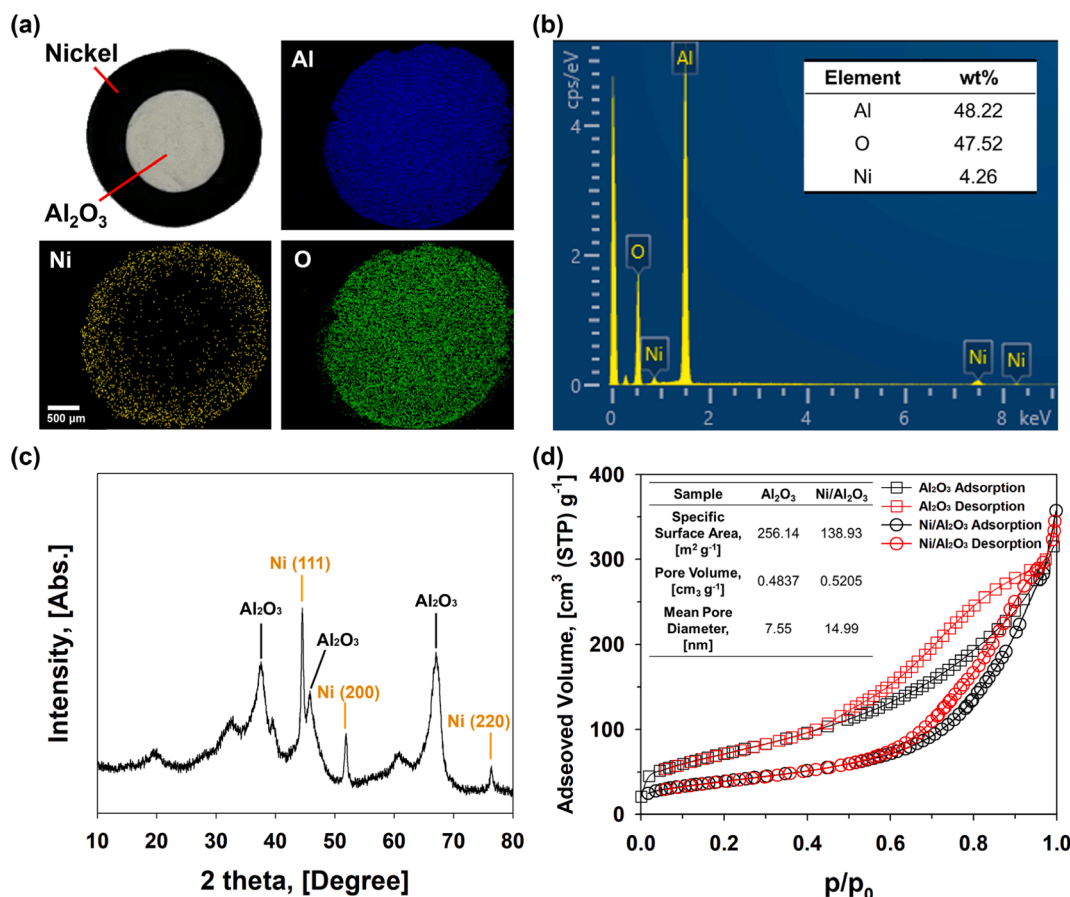


Fig. 1. (a) Cross-section SEM image and inorganic distribution, (b) elemental composition obtained from EDX test, (c) XRD analysis results, and (d) N<sub>2</sub>-adsorption/desorption patterns of the prepared catalyst (Ni/Al<sub>2</sub>O<sub>3</sub>).

particularly aimed at transforming into value-added products in the pyrolysis process. In pursuit of improving a sustainability to the pyrolysis system, this study employed CO<sub>2</sub>, a greenhouse gas, as a reactive agent. Considering the thermodynamic stability of CO<sub>2</sub>, its utilization in the pyrolysis process could be technically challenging. Nevertheless, introducing CO<sub>2</sub> in the pyrolysis process presents a promising approach for mitigating greenhouse gas emissions and improving carbon utilization. Therefore, this study placed significant importance on elucidating the mechanistic role of CO<sub>2</sub> in enhancing syngas production, highlighting its unique ability as a soft oxidant that supports the thermal cracking of biocrude [42]. To enhance the effectiveness of CO<sub>2</sub>, a nickel (Ni)-based catalyst was introduced to facilitate the enhanced cleaving of C–C, C–O, and C–H bonds in biocrude [43]. To elucidate the functionality of CO<sub>2</sub>, qualitative and quantitative analyses were conducted on the gaseous and liquid pyrogenic products derived from RTR under CO<sub>2</sub> conditions, and the results were compared with those obtained under nitrogen (N<sub>2</sub>) conditions. This comparative approach provided valuable insights into the impact of CO<sub>2</sub> as a partial oxidant and the catalytic influence on the pyrolysis of RTR. These findings contribute to a deeper understanding of the dynamic processes involved and offer potential avenues for optimizing sustainable syngas production.

## 2. Materials and methods

### 2.1. Sample preparation and chemical reagents

RTR was sourced from the National Institute of Horticultural and Herbal Science in Korea. The RTR was initially chopped and then pre-treated in a convection oven, isothermally maintained at 105 °C, to remove moisture over a 24-hour period. Then the dried RTR was pulverized and sieved through a 50-mesh sieve to ensure a uniform particle size ( $\leq 300 \mu\text{m}$ ). Calibration for inductively coupled plasma optical emission spectroscopy (ICP-OES) was conducted using an inorganic solution containing 21 elements at a concentration of 1000 ppm each (purchased from Merck, Hesse, Germany). Activated aluminum oxide (Al<sub>2</sub>O<sub>3</sub>, approximately 3 mm diameter, 320–360 m<sup>2</sup>g<sup>−1</sup> specific surface area; Prod. # 1038-1405), methanol (CH<sub>3</sub>OH, 99.5 %, Prod. # 5558-4410), dichloromethane (CH<sub>2</sub>Cl<sub>2</sub>, 99.8 %, Prod. # 3030-2304), nickel (II) nitrate hexahydrate (Ni(NO<sub>3</sub>)<sub>2</sub>·6H<sub>2</sub>O, 97 %, Prod. # 5608-4405), and 1-octanol (C<sub>8</sub>H<sub>17</sub>OH,  $\geq 98$  %, Prod. # 5634-4405) were purchased from Daejung Chemicals (Korea). High-purity grade gases (99.999 vol%) including air, N<sub>2</sub>, and CO<sub>2</sub> were sourced from Seoul Specialty Gases (Korea). For a micro-GC calibration, a gas mixture containing N<sub>2</sub>, CO<sub>2</sub>, H<sub>2</sub>, CO, and CH<sub>4</sub> was obtained from Union Gas (Korea).

### 2.2. Catalyst preparation

In the catalytic pyrolysis of RTR, an eggshell-type Ni catalyst with a thin layer of active metal sites was synthesized, following methods described elsewhere [44]. Preparation involved drying Al<sub>2</sub>O<sub>3</sub> support material at 500 °C for 3 h under N<sub>2</sub> condition, followed by immersion in hydrophobic solvent (1-octanol) and sonication for 3 h to fill the support's interior with the 1-octanol solution. After filtering any remaining octanol, a thin active site layer was formed by dissolving the prepared nickel precursor solution, equivalent to 5 wt% Ni on the Al<sub>2</sub>O<sub>3</sub> support. The nickel solution was dripped onto the support at 140 °C, fixing the Ni-precursor on the surface as the hydrophilic solvent evaporated. The temperature was then raised to 200 °C for 2 h to complete the evaporation of the hydrophobic solvent. The dried Al<sub>2</sub>O<sub>3</sub> was then calcined at 800 °C for 3 h, followed by a reduction of the calcined Ni-impregnated Al<sub>2</sub>O<sub>3</sub> using a 20 vol% mixture of H<sub>2</sub> gas balanced with N<sub>2</sub> at 800 °C for 2 h. The hydrogenation temperature of nickel oxide on Al<sub>2</sub>O<sub>3</sub> is higher than 750 °C [45]. To exhibit the distribution of both organic and inorganic particles on the eggshell-type structure, a cross-section of Ni/Al<sub>2</sub>O<sub>3</sub> was scanned using scanning electron microscopy/energy

**Table 1**

Ultimate, proximate, and ICP-OES analyses of RTR.

|                                       |       |
|---------------------------------------|-------|
| Elemental analysis (wt% on dry basis) |       |
| Carbon (C)                            | 48.58 |
| Proximate analysis (wt% on dry basis) |       |
| Hydrogen (H)                          | 6.09  |
| Oxygen (O)                            | 41.36 |
| Nitrogen (N)                          | 1.09  |
| Moisture                              | 3.68  |
| Volatile matter                       | 70.09 |
| Fixed carbon                          | 23.42 |
| Ash                                   | 2.81  |
| ICP-OES analysis (wt% on dry basis)   |       |
| Sodium (Na)                           | 0.03  |
| Potassium (K)                         | 1.13  |
| Magnesium (Mg)                        | 0.16  |
| Calcium (Ca)                          | 0.07  |

dispersive X-ray spectroscopy (SEM/EDX, Cube-II Mini-SEM, Emcraft, Gyeonggi-do, Korea). Fig. 1(a) shows that the dispersal of Ni particles on the outer layer of Al<sub>2</sub>O<sub>3</sub> was concentrated. The surface elements on cross-section of the prepared catalyst were quantified by testing EDX analysis (Fig. 1(b)). To measure the precise loading of Ni, the Ni/Al<sub>2</sub>O<sub>3</sub> was analyzed using ICP-OES test, determining a Ni mass of 4.92 wt%. The crystallinity of the Ni/Al<sub>2</sub>O<sub>3</sub> was examined using an X-ray diffraction (XRD, D8 Advance, Bruker, USA) in the angular range of 10° to 80° (Fig. 1(c)). To evaluate the N<sub>2</sub>-physisorption capability, Brunauer–Emmett–Teller/Barrett–Joyner–Halenda (BET/BJH) analysis was conducted, determining specific surface area, pore volume, and pore size distribution of the alumina support (Al<sub>2</sub>O<sub>3</sub>) and the prepared catalyst (Ni/Al<sub>2</sub>O<sub>3</sub>), as presented in Fig. 1(d).

### 2.3. Characterization of RTR

Thermogravimetric analysis (TGA) of RTR was conducted to determine its thermal mass degradation patterns under N<sub>2</sub> and CO<sub>2</sub> conditions, using an STA 449 F5 instrument (NETZSCH, Germany). The temperature increased from 35 °C to 900 °C at a rate of 10 °C min<sup>−1</sup>, with a flow rate set at 100 mL min<sup>−1</sup>. Proximate analysis, as per ASTM D7582, was also performed using TGA. The moisture content in RTR was determined by measuring weight loss after drying at 105 °C for 1 h in an N<sub>2</sub> atmosphere. Volatile matter was quantified by subjecting RTR to carbonization at 925 °C for 7 min under N<sub>2</sub> conditions. Ash content was determined by oxidizing at 800 °C until a constant weight was achieved. The elemental analyzer (EA, EA3100, EuroVector, Lombardy, Italy) was used to ascertain mass percentages of organic elements in RTR. Inorganic elements in RTR were assessed using an ICP-OES unit (5800, Agilent Technologies, California, USA). Comprehensive results from the ultimate analysis, proximate analysis, and ICP-OES analysis are presented in Table 1.

### 2.4. Pyrolysis of RTR

Three distinct pyrolysis setups (single-stage, multi-stage, and catalytic pyrolysis) were used to convert RTR into the pyrogenic products outlined in [46]. A schematic of the three pyrolysis setups employed in this study is presented in Fig. S1. Each pyrolysis was carried out in a quartz tubular reactor (TR) with 22 mm diameter and 1000 mm length. The use of a quartz TR was intended to prevent catalytic interactions. Both ends of the reactor were sealed with vacuum fittings, employing a stainless-steel union (SS-16-UT-6, Swagelok, USA) and reducer (SS-1610-6-4, Swagelok, USA). In the single-stage pyrolysis process, the TR was enclosed in a tubular furnace (FT-400, DAIHAN Scientific, Korea), and the temperature was raised from 100 °C to 700 °C at a rate of 10 °C min<sup>−1</sup>. For multi-stage pyrolysis, an additional (second) tubular furnace, set at 600 °C, was added next to the existing (first) furnace. In catalytic pyrolysis, mirroring the multi-stage method, a Ni/Al<sub>2</sub>O<sub>3</sub> catalyst was

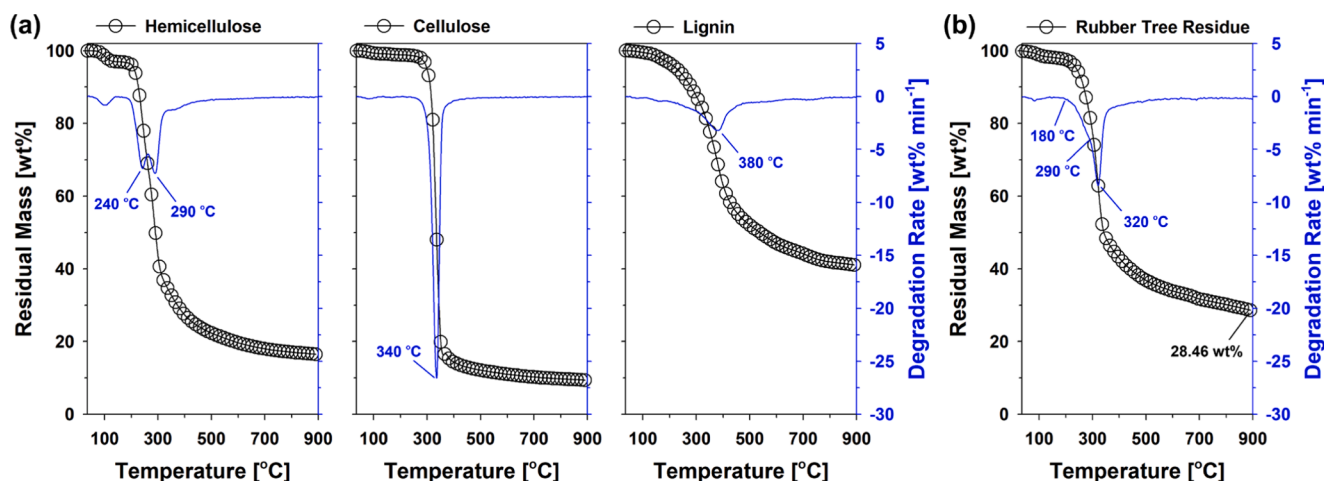


Fig. 2. Mass change (black) and degradation rate (blue) of (a) hemicellulose, cellulose, and lignin, (b) RTR under  $N_2$ .

introduced in the middle of the second furnace, at a constant temperature of 600 °C. Each experiment had  $10.00 \pm 0.01$  g of RTR loaded inside the TR located in the middle of the first furnace. A series of reference pyrolysis tests were conducted under a continuous flow of  $800 \text{ mL min}^{-1}$  of  $N_2$  gas, while  $CO_2$ -assisted pyrolysis tests involved introducing 80 vol %  $CO_2$  and 20 vol %  $N_2$ . All the experiments were triplicated to ensure accuracy and precision. Standard deviations of three experimental results were less than 5 %. All the experimental results indicate average values from three replicates, and their standard deviations are less than 5 %.

## 2.5. Analysis of pyrogenic products

Condensable pyrolysates kept at  $-40$  °C were initially diluted with methanol (100 mL) and dichloromethane (100 mL). The use of mixed solution with methanol and dichloromethane facilitated dissolving both hydrophilic and hydrophobic chemicals trapped in the condensation unit. The condensable ones were analyzed with a gas chromatography-mass spectrometer (GC-MS, 8890, Agilent Technologies, California, USA) equipped with a DB-WAX column ( $30 \text{ m} \times 0.25 \text{ mm} \times 0.25 \text{ }\mu\text{m}$ ) to identify chemical compounds. For non-condensable gaseous products, a micro-GC system (990, Agilent Technologies, California, USA) was used, featuring a thermal conductivity detector (TCD) equipped with two columns: molecular sieve (5A,  $10 \text{ m} \times 320 \text{ }\mu\text{m} \times 30 \text{ }\mu\text{m}$ ) and capillary column (Plot U,  $8 \text{ m} \times 320 \text{ }\mu\text{m} \times 30 \text{ }\mu\text{m}$ ).

## 3. Results and discussion

### 3.1. Characterization of thermal behaviors of RTR

The expansion of rubber plantations is ongoing despite the commercial lifespan of a single rubber tree being limited to 30 years [41]. This has led to the massive generation of RTR, emphasizing the need for its sustainable valorisation. The rubber tree is a lignocellulose species [47] and primarily consists of the three key constituents such as hemicellulose, cellulose, and lignin [48]. Latex content in the tree is less than 2 wt% [49]. However, due to the distinctive characteristics of the rubber tree in latex production [50], the thermolytic behaviors of RTR may differ from those of traditional LB. To investigate this, thermal degradations of the three constituents enriched in lignocellulose were conducted using the TGA instrument under a  $N_2$  atmosphere. Mass changes of each constituent and their degradation rates were plotted from 25 °C to 900 °C at a constant rate of  $10 \text{ }^\circ\text{C min}^{-1}$ , as shown in Fig. 2(a). The TGA results for RTR under the  $N_2$  environment were also plotted in an identical temperature program in Fig. 2(b).

As depicted in Fig. 2(a), thermal degradation profiles of

hemicellulose, cellulose, and lignin exhibited distinctive thermal characteristics. Major mass reductions associated with the individual constituents occurred at different temperature windows, reflecting the diverse thermolytic behaviors linked to variations in their inherent chemical structures [51]. Hemicellulose, for example, initiated its thermolysis at 180 °C, manifesting two peak degradation rates at 240 °C and 290 °C. The presence of two characteristic degradation rates suggests that hemicellulose comprises a polymeric backbone containing at least two different sugar compounds. This observation aligns with the known branched structure of hemicellulose, which is a polymer composed of various sugars [52]. Cellulose exhibited mass reduction in a narrower temperature range, specifically between 250 °C and 400 °C, with a fast degradation rate observed at 340 °C. The rapid thermolysis of cellulose is attributed to bond scissions occurring within the repeated glucose backbone [53].

In contrast to the TGA results from hemicellulose and cellulose, the major thermal degradation of lignin occurred over a much broader temperature range (180–500 °C), with the fastest thermolytic rate observed at 380 °C. Lignin, characterised by complex matrices intertwined with several phenolic analogs [54], exhibited superior thermal stability, influenced by the resonance nature of these structures, leading to significant retardations in thermolysis [55,56]. The degradation rate became notably slow at temperatures  $\geq 500$  °C, indicating the recombination of phenolic compounds instead of their volatilisation [57]. In addition, dehydrogenation, occurring proportionally to the reaction temperature, provides favorable conditions for biochar formation [58]. Although similar trends were identified from the thermolysis of hemicellulose and cellulose, the yield of biochar formation stemming from lignin was much larger than that originating from (hemi)cellulose.

The thermolytic profiles for RTR under the  $N_2$  environment are shown in Fig. 1(b). The major mass changes of RTR occurred at  $\leq 500$  °C, accounting for 61.05 wt%. The peak degradation rate appeared at 320 °C. Considering the similarity with the temperature (340 °C) indicating the peak degradation rate of cellulose, the volatilisation of RTR is mainly derived from the bond scissions of the glucose-based backbone in cellulose [53]. The marginal temperature difference may be attributed to variations in the degree of polymerisation of the cellulose used in the TGA test [59]. Of note, an increase in the molecular moiety of the carbonaceous feedstock potentially retards the thermal transfer in the pyrolysis process [60]. The temperature initiating the thermal degradation of RTR (180 °C) was lower than that of the cellulose (250 °C). Regardless, the characteristic degradation rates identified from the TGA test of hemicellulose were not detected. These thermolytic patterns demonstrate a trace existence of hemicellulose constituent in RTR.

Interestingly, the final residual mass of RTR at 900 °C was 28.46 wt



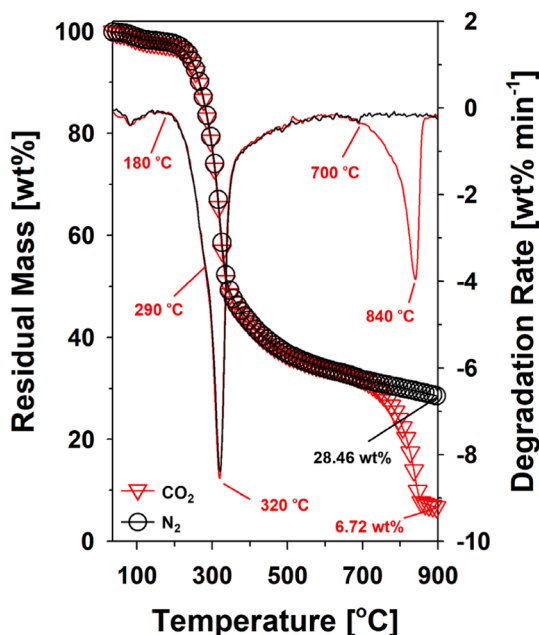


Fig. 3. Mass change and degradation rate of RTR under the N<sub>2</sub> (black) and CO<sub>2</sub> (red) environments.

%, which was much higher than those from (hemi)cellulose, suggesting that RTR comprises a significant proportion of lignin. This could be closely related to the plantation environment of the rubber tree. The tree is preferentially cultivated in tropical regions [61], facing harsh climates such as high temperatures, intense sunlight, and frequent rainfall [62]. In response to these challenging environments, the rubber tree has evolved dense structural formations [63]. In addition, the presence of latex in the rubber tree aids the cell growth of lignocellulosic components with improved resistance to spoilages, pests, and physical stresses [64]. Thus, the TGA results for RTR indicate well the characteristic composition of hemicellulose, cellulose, and lignin.

To investigate the efficacy of CO<sub>2</sub> from the thermolysis of RTR, TGA tests were conducted under a CO<sub>2</sub> environment. The mass change of RTR and its degradation rate under the CO<sub>2</sub> environment were evaluated at programmed temperatures ranging from 25 °C to 900 °C at a constant rate of 10 °C min<sup>-1</sup>. To elucidate the effectiveness of CO<sub>2</sub>, the TGA results from the CO<sub>2</sub> environment were compared with those obtained under a N<sub>2</sub> environment (Fig. 3).

A noted observation in Fig. 3 was that the additional thermal degradation of RTR only occurred under the CO<sub>2</sub> condition at temperatures  $\geq 700$  °C. The final residual mass of RTR at 900 °C under the CO<sub>2</sub> environment was 6.72 wt%, which was significantly lower than that under the N<sub>2</sub> environment (28.46 wt%). This difference signifies the participation of CO<sub>2</sub> in the heterogeneous reaction with the residual carbon from the thermolysis of RTR at  $\geq 700$  °C. This is likely due to the occurrence of the Boudouard reaction (BR,  $C_{(s)} + CO_{2(g)} \rightleftharpoons 2CO_{(g)}$ ) [65], as the thermo-chemical dynamic calculation on the BR suggests its spontaneous activation at 710 °C [66]. However, the initiation temperature for the BR in the thermolysis of RTR was observed as being 10 °C lower. This suggests that the inorganic components in RTR may have catalysed the BR [67]. As indicated in Table 1, RTR contains a notable portion of alkaline metals such as K, which could potentially accelerate the starting temperature for the BR [68]. Regardless, the final residual mass (6.72 wt%) under the CO<sub>2</sub> condition differed from the ash content (2.81 wt%) after proximate analysis (Table 1). This difference suggests that the BR induced by CO<sub>2</sub> was not fully activated due to the slow reaction kinetics. At the same time, the trends in the mass changes from the thermolysis of RTR at temperatures  $\leq 700$  °C were identical under the N<sub>2</sub> and CO<sub>2</sub> environments. This indicates that CO<sub>2</sub> is unable to activate the heterogeneous reaction with RTR due to insufficient energy supply. Regardless, the occurrence of the homogeneous reaction (i.e., gas-phase reaction (GPR)) of CO<sub>2</sub> with VMs liberated from the thermolysis of RTR at  $\leq 700$  °C remains unclear. It is worth noting that the TGA test only provides the thermo-gravimetric results of RTR [69].

### 3.2. Pyrolysis of RTR under CO<sub>2</sub> atmosphere

To investigate the mechanistic functions of CO<sub>2</sub> via the gas-phased homogeneous reaction, a lab-scale pyrolysis test of RTR was performed under N<sub>2</sub> and CO<sub>2</sub> atmospheres. RTR (with a loading of  $10.00 \pm 0.01$  g) was thermally degraded at programmed temperatures (100 °C to 700 °C at a constant rate of 10 °C min<sup>-1</sup>). Major gaseous products derived from RTR were measured as a function of reaction temperature. In Fig. 4(a), the concentration curves for the pyrogenic gases from the CO<sub>2</sub> atmosphere were presented in comparison to the results from the N<sub>2</sub> atmosphere. Error bars representing the standard deviations for all the concentration data were incorporated in Fig. 4(a). Mass balances of three pyrogenic products liberated from the pyrolysis of RTR under both atmospheres are illustrated in Fig. 4(b).

In Fig. 4(a), the evolutionary trends for H<sub>2</sub>, CO, and CH<sub>4</sub> are linked to the pyrolysis test temperature, influencing the chemical bond scission of key polymeric components in RTR. The generation of CO at the

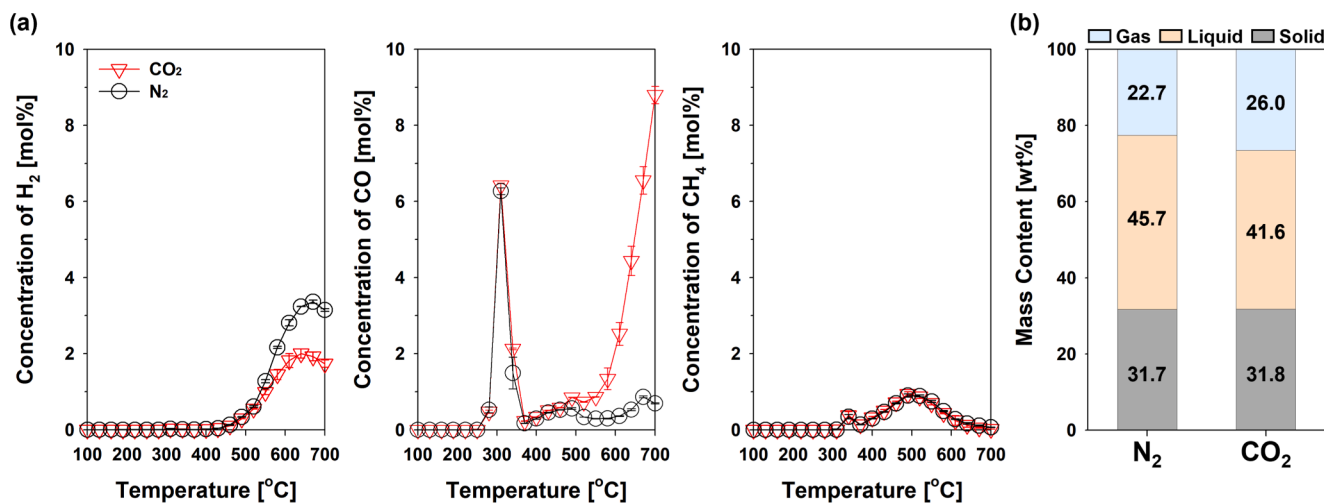


Fig. 4. (a) Concentration curves of major pyrogenic gases and (b) mass balances of three pyrogenic products (gas, liquid, and solid) from the pyrolysis of RTR under the N<sub>2</sub> and CO<sub>2</sub> atmospheres. The experimental results indicate average values from three replicates, and their standard deviations are less than 5%.

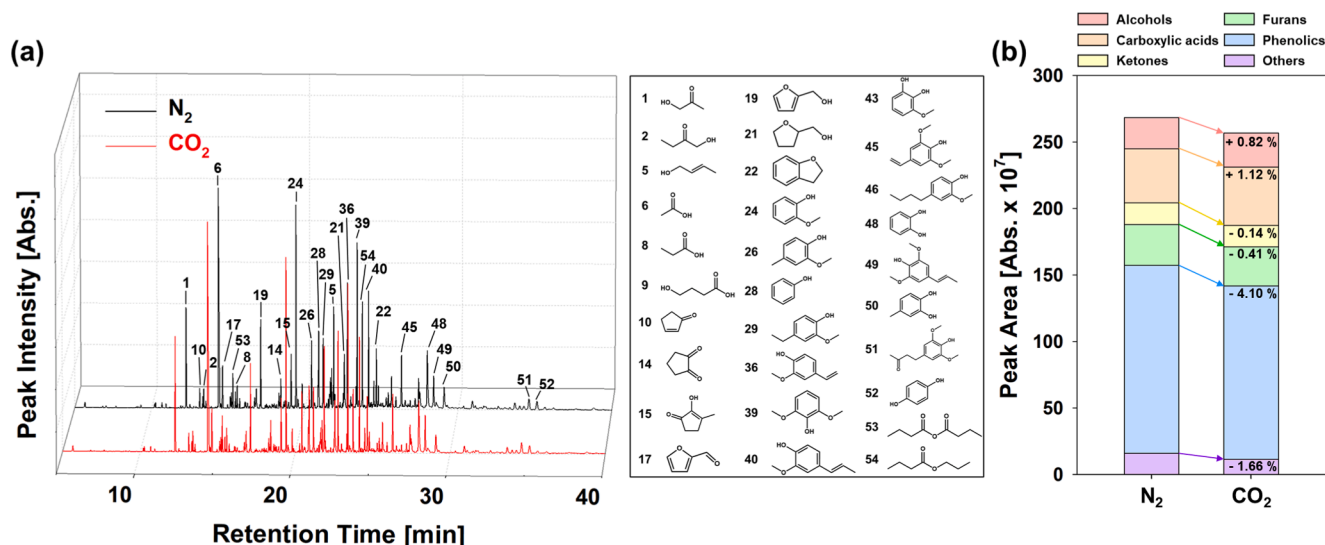


Fig. 5. (a) GC/MS chromatograms, and (b) total peak areas of representative chemical species in the biocrude derived from pyrolysis of RTR under the N<sub>2</sub> and CO<sub>2</sub> atmospheres.

temperature range of 280–370 °C was initiated first, developing a peak concentration at 310 °C. The evolution patterns of CO were likely attributed to deoxygenation of the polymeric backbone in (hemi)cellulose, given their abundance in RTR [70]. The temperature window showing the CO evolution corresponds well with the TGA result indicating the major volatilisation of (hemi)cellulose in RTR (Fig. 1(b)). The concentration of CH<sub>4</sub> at 300–700 °C was observed, reaching the highest concentration level at 500 °C. The CH<sub>4</sub> formation results from the thermal dissociation of methyl and/or methoxy linkages enriched in the phenolic analogs [71]. Their depletion at ≥ 500 °C resulted in a decrease in CH<sub>4</sub> concentration. The evolution of H<sub>2</sub> developed at the higher temperature window (400–700 °C), with the highest concentration level at 670 °C. The continuous increase of H<sub>2</sub> up to 670 °C is reasonable considering that the degree of dehydrogenation increases proportionally with the reaction temperature [72].

Under the CO<sub>2</sub> condition, the concentration curves of major pyrogenic gases at ≤ 500 °C were nearly identical to the results from the N<sub>2</sub> condition. However, the gas evolution patterns between the N<sub>2</sub> and CO<sub>2</sub> conditions were different at ≥ 500 °C. Increased formation of CO was identified under the CO<sub>2</sub> condition. Simultaneous suppression of H<sub>2</sub> was observed within the same temperature window. The CO enhancement induced by CO<sub>2</sub> can be expected from well-defined chemical reactions such as the reverse water–gas shift reaction (rWGS, H<sub>2</sub> + CO<sub>2</sub> ⇌ CO + H<sub>2</sub>O) [73]. In the rWGS reaction, H<sub>2</sub> is consumed to stoichiometrically produce CO in the presence of CO<sub>2</sub> [74]. Nevertheless, the enhanced formation of CO exceeded the decreased formation of H<sub>2</sub>. Therefore, the surplus evolution of CO cannot be explained without considering an additional carbon and/or oxygen source. However, the enhancement of CO appearing at ≤ 700 °C excludes any activation of the BR, proposing that a potential occurrence of the gas-phased homogeneous reaction of CO<sub>2</sub> with the VMs derived from RTR exists. In short, CO<sub>2</sub> has the opportunity to be a reactive agent for the production of CO. The GPR between CO<sub>2</sub> and VMs could alter the compositional matrices of the pyrogenic products. This hypothesis was well supported from mass balances of the three pyrogenic products (Fig. 4(b)). The mechanistic role of CO<sub>2</sub> allowed for the conversion of biocrude into gaseous products.

To assess the functionality of CO<sub>2</sub> on compositional modification, the biocrudes stemming from RTR under N<sub>2</sub> and CO<sub>2</sub> atmospheres were qualitatively analyzed using GC/MS instrumentation. The resulting chromatograms of the biocrudes are presented in Fig. 5(a). Major chemical constituents in the chromatograms are labelled, and their

chemical structures are visualised. The chemical constituents can be classified into specific functional linkages (alcohols, carboxylic acids, ketones, furans, phenolics, and others), and their total peak areas are presented in Fig. 5(b). The qualitative information on the identified chemical constituents is summarised in Table 2.

The compositional matrices in the biocrude were notably influenced by the chemical composition of (hemi)cellulose and lignin in RTR. The biocrude derived from RTR was dominated by phenolic compounds. The abundance of phenolics is likely attributed to the high lignin content in RTR, as confirmed in Fig. 2(b) and Fig. 4(a). The biocrude also contains alcohol, carboxylic acids, ketones, and furans. Their formations are derived from the thermolytic bond scissions of the polymeric backbone in (hemi)cellulose [75]. Overall chromatogram trends of the biocrudes from the N<sub>2</sub> and CO<sub>2</sub> atmospheres are nearly identical. This signifies that CO<sub>2</sub> does not affect the change in the chemical species. Nevertheless, the use of CO<sub>2</sub> resulted in a slightly decreased formation of the biocrude, indicating a CO<sub>2</sub>:N<sub>2</sub> ratio of 0.96 in the total peak areas. This suggests that CO<sub>2</sub> enhances the thermal decomposition of VMs derived from RTR. However, the effectiveness of CO<sub>2</sub> via the GPR was susceptible to the functional linkages of the chemical constituents. Specifically, the introduction of CO<sub>2</sub> allowed for 4.07 and 0.4 area% reductions in the phenolic and furanic compounds, respectively, compared to the results from the N<sub>2</sub> atmosphere. The functional reactivity of CO<sub>2</sub> for the GPR with the VMs and the simultaneous CO production were effective in the phenolic compounds. This proposes that CO<sub>2</sub> has great potential to break down aromatic compounds, despite their robust nature [76,77]. Interestingly, the formation of alcohols and carboxylic acids increased by 0.82 and 1.11 area%, respectively. The slight increase in the specific chemical groups is likely a result of the enhanced conversion of the VM induced by CO<sub>2</sub> into shorter molecular compounds.

### 3.3. Multi-stage and catalytic pyrolysis of RTR under CO<sub>2</sub> atmosphere

As previously discussed, the homogeneous GPR induced by CO<sub>2</sub> resulted in the conversion of biocrude into gaseous products. However, the efficacy of CO<sub>2</sub> was limited at temperatures ≥ 500 °C, despite the major liberation of VMs occurring at ≤ 500 °C. To identify the functional reactivity of CO<sub>2</sub> in the entire temperature window, the experimental setups for the pyrolysis tests were modified by adding an isothermal heating zone maintained at 600 °C, affecting multi-stage pyrolysis. Note that selecting the isothermal temperature of 600 °C in the multi-step pyrolysis is strongly associated with the temperature regime

**Table 2**Peak area and its ratio of the major chemical constituents in the biocrudes obtained from the pyrolysis of RTR under N<sub>2</sub> and CO<sub>2</sub> conditions.

| Order                      | R.T.,<br>[min] | Chemical constituents                               | Area in N <sub>2</sub> ,<br>[Abs. × 10 <sup>6</sup> ] | Area in CO <sub>2</sub> ,<br>[Abs. × 10 <sup>6</sup> ] | Ratio of CO <sub>2</sub> to N <sub>2</sub> |
|----------------------------|----------------|-----------------------------------------------------|-------------------------------------------------------|--------------------------------------------------------|--------------------------------------------|
| <i>1. Alcohols</i>         |                |                                                     |                                                       |                                                        |                                            |
| 1                          | 12.27          | 1-hydroxy-2-propanone                               | 94.46                                                 | 108.48                                                 | 1.15                                       |
| 2                          | 13.41          | 1-hydroxy-2-butanone                                | 17.09                                                 | 19.76                                                  | 1.16                                       |
| 3                          | 13.54          | 1-methyl-2-hydroxyacetate                           | 6.57                                                  | 6.81                                                   | 1.04                                       |
| 4                          | 16.74          | 1,2-ethanediol                                      | 17.79                                                 | 16.59                                                  | 0.93                                       |
| 5                          | 21.88          | <i>trans</i> -2-butenol                             | 98.93                                                 | 105.31                                                 | 1.06                                       |
|                            |                | Subtotal                                            | 234.84                                                | 256.95                                                 | 1.09                                       |
| <i>2. Carboxylic acids</i> |                |                                                     |                                                       |                                                        |                                            |
| 6                          | 14.37          | acetic acid                                         | 338.46                                                | 370.49                                                 | 1.09                                       |
| 7                          | 15.14          | formic acid                                         | 18.76                                                 | 16.46                                                  | 0.88                                       |
| 8                          | 15.59          | propanoic acid                                      | 22.56                                                 | 22.27                                                  | 0.99                                       |
| 9                          | 16.84          | 4-hydroxy-butanolic acid                            | 27.57                                                 | 28.07                                                  | 1.02                                       |
|                            |                | Subtotal                                            | 407.35                                                | 437.30                                                 | 1.07                                       |
| <i>3. Ketones</i>          |                |                                                     |                                                       |                                                        |                                            |
| 10                         | 13.15          | 2-cyclopenten-1-one                                 | 20.01                                                 | 19.68                                                  | 0.98                                       |
| 11                         | 13.34          | 2-methyl-2-cyclopenten-1-one                        | 11.03                                                 | 10.37                                                  | 0.94                                       |
| 12                         | 15.47          | 3-methyl-2-cyclopenten-1-one                        | 16.44                                                 | 15.02                                                  | 0.91                                       |
| 13                         | 15.75          | 2,3-dimethyl-2-cyclopenten-1-one                    | 9.76                                                  | 7.89                                                   | 0.81                                       |
| 14                         | 18.44          | 1,2-cyclopentanedione                               | 27.35                                                 | 31.79                                                  | 1.16                                       |
| 15                         | 19.11          | 2-hydroxy-3-methyl-2-cyclopenten-1-one              | 55.78                                                 | 53.15                                                  | 0.95                                       |
| 16                         | 19.82          | 3-ethyl-2-hydroxy-2-cyclopenten-1-one               | 23.80                                                 | 22.54                                                  | 0.95                                       |
|                            |                | Subtotal                                            | 164.18                                                | 160.45                                                 | 0.98                                       |
| <i>4. Furans</i>           |                |                                                     |                                                       |                                                        |                                            |
| 17                         | 14.64          | furfural                                            | 47.64                                                 | 47.91                                                  | 1.01                                       |
| 18                         | 15.22          | 1-(2-furanyl)-ethanone                              | 14.58                                                 | 13.91                                                  | 0.95                                       |
| 19                         | 17.13          | 2-furanmethanol                                     | 73.02                                                 | 74.20                                                  | 1.02                                       |
| 20                         | 18.31          | 2(5 <i>H</i> )-furanone                             | 13.30                                                 | 14.94                                                  | 1.12                                       |
| 21                         | 22.57          | tetrahydro-2-furanmethanol                          | 84.85                                                 | 74.88                                                  | 0.88                                       |
| 22                         | 24.66          | 2,3-dihydro-benzofuran                              | 55.15                                                 | 53.44                                                  | 0.97                                       |
| 23                         | 25.44          | 5-(hydroxymethyl)dihydrofuran-2(3 <i>H</i> )-one    | 17.10                                                 | 15.45                                                  | 0.90                                       |
|                            |                | Subtotal                                            | 305.64                                                | 294.74                                                 | 0.96                                       |
| <i>5. Phenolics</i>        |                |                                                     |                                                       |                                                        |                                            |
| 24                         | 19.43          | 2-methoxy-phenol                                    | 170.59                                                | 161.54                                                 | 0.95                                       |
| 25                         | 20.30          | 2-methoxy-5-methyl-phenol                           | 10.70                                                 | 8.99                                                   | 0.84                                       |
| 26                         | 20.44          | 2-methoxy-4-methyl-phenol                           | 54.86                                                 | 49.59                                                  | 0.90                                       |
| 27                         | 20.87          | 2-methyl-phenol                                     | 33.56                                                 | 28.64                                                  | 0.85                                       |
| 28                         | 20.91          | phenol                                              | 64.63                                                 | 59.33                                                  | 0.92                                       |
| 29                         | 21.19          | 4-ethyl-2-methoxy-phenol                            | 62.97                                                 | 59.86                                                  | 0.95                                       |
| 30                         | 21.53          | 2-ethyl-phenol                                      | 8.94                                                  | 7.73                                                   | 0.87                                       |
| 31                         | 21.61          | 2,6-dimethyl-phenol                                 | 13.31                                                 | 10.50                                                  | 0.79                                       |
| 32                         | 21.66          | 4-methyl-phenol                                     | 41.07                                                 | 33.48                                                  | 0.82                                       |
| 33                         | 21.74          | 3-methyl-phenol                                     | 33.67                                                 | 27.45                                                  | 0.82                                       |
| 34                         | 21.96          | 2-methoxy-4-propyl-phenol                           | 19.39                                                 | 15.90                                                  | 0.82                                       |
| 35                         | 22.51          | 2-methoxy-4-(2-propenyl)-phenol                     | 14.07                                                 | 13.14                                                  | 0.93                                       |
| 36                         | 22.78          | 2-methoxy-4-vinylphenol                             | 105.55                                                | 103.19                                                 | 0.98                                       |
| 37                         | 23.00          | 3,4-dimethyl-phenol                                 | 10.45                                                 | 8.23                                                   | 0.79                                       |
| 38                         | 23.34          | 2-methoxy-4-(1-propenyl)-phenol                     | 19.21                                                 | 17.12                                                  | 0.89                                       |
| 39                         | 23.41          | 2,6-dimethoxy-phenol                                | 141.52                                                | 136.91                                                 | 0.97                                       |
| 40                         | 24.16          | 2-methoxy-4-(1-propenyl)-phenol                     | 106.97                                                | 103.30                                                 | 0.97                                       |
| 41                         | 24.50          | 4-vinylphenol                                       | 24.63                                                 | 22.65                                                  | 0.92                                       |
| 42                         | 25.31          | 2,6-dimethoxy-4-propylphenol                        | 18.63                                                 | 13.21                                                  | 0.71                                       |
| 43                         | 25.65          | 3-methoxy-1,2-benzenediol                           | 33.62                                                 | 32.87                                                  | 0.98                                       |
| 44                         | 25.94          | 2,6-dimethoxy-4-(2-propenyl)-Phenol                 | 11.35                                                 | 10.25                                                  | 0.90                                       |
| 45                         | 26.30          | 4-ethenyl-2,6-dimethoxy-phenol                      | 63.20                                                 | 67.64                                                  | 1.07                                       |
| 46                         | 27.42          | 4-butyl-2-methoxy-Phenol                            | 38.29                                                 | 37.05                                                  | 0.97                                       |
| 47                         | 27.49          | 1-(2,3,4-trihydroxyphenyl)-ethanone                 | 35.46                                                 | 28.44                                                  | 0.80                                       |
| 48                         | 27.99          | 1,2-dihydroxybenzene                                | 125.42                                                | 99.40                                                  | 0.79                                       |
| 49                         | 28.40          | ( <i>E</i> )-2,6-dimethoxy-4-(prop-1-en-1-yl)phenol | 48.79                                                 | 53.21                                                  | 1.09                                       |
| 50                         | 29.09          | 4-methyl-1,2-benzenediol                            | 44.37                                                 | 38.66                                                  | 0.87                                       |
| 51                         | 34.61          | 4-(p-methoxyphenyl)-2-butanone                      | 30.54                                                 | 29.90                                                  | 0.98                                       |
| 52                         | 35.12          | 1,4-benzenediol                                     | 27.40                                                 | 24.92                                                  | 0.91                                       |
|                            |                | Subtotal                                            | 1,413.14                                              | 1,303.09                                               | 0.92                                       |
| <i>6. Others</i>           |                |                                                     |                                                       |                                                        |                                            |
| 53                         | 15.31          | n-butyric anhydride                                 | 32.04                                                 | 34.97                                                  | 0.92                                       |
| 54                         | 23.74          | n-propyl butyrate                                   | 101.40                                                | 60.59                                                  | 0.60                                       |
| 55                         | 24.79          | 3-pyridinol                                         | 25.79                                                 | 19.03                                                  | 0.74                                       |
|                            |                | Subtotal                                            | 159.23                                                | 114.59                                                 | 0.72                                       |
|                            |                | Total                                               | 2,684.39                                              | 2,567.11                                               | 0.96                                       |

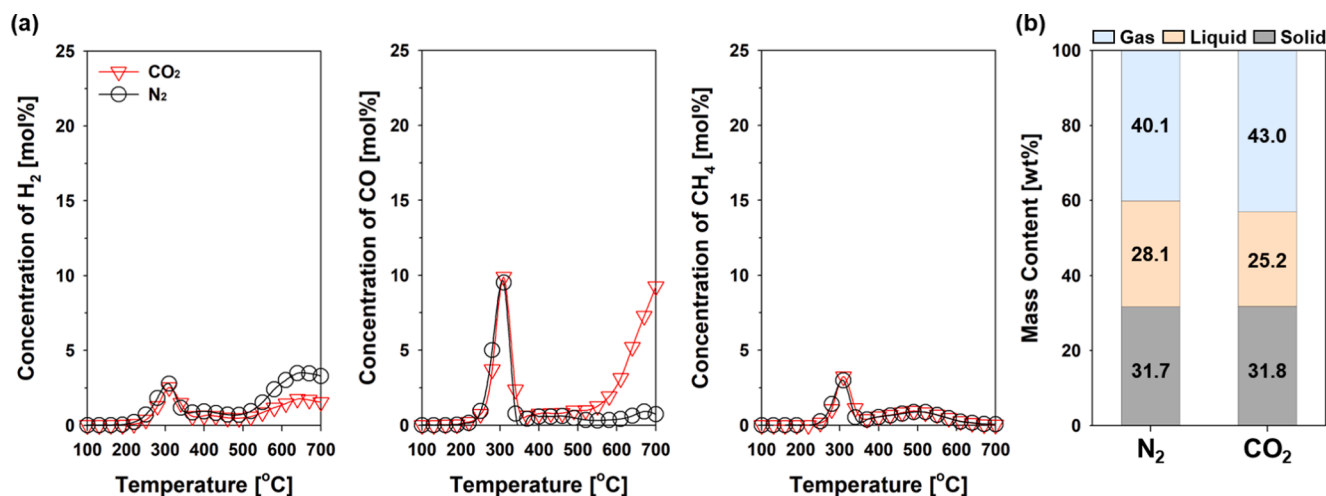


Fig. 6. (a) Concentration curves of major pyrogenic gases, and (b) mass balances of three pyrogenic products (gas, liquid, and solid) from the multi-stage pyrolysis of RTR under the N<sub>2</sub> and CO<sub>2</sub> atmospheres.

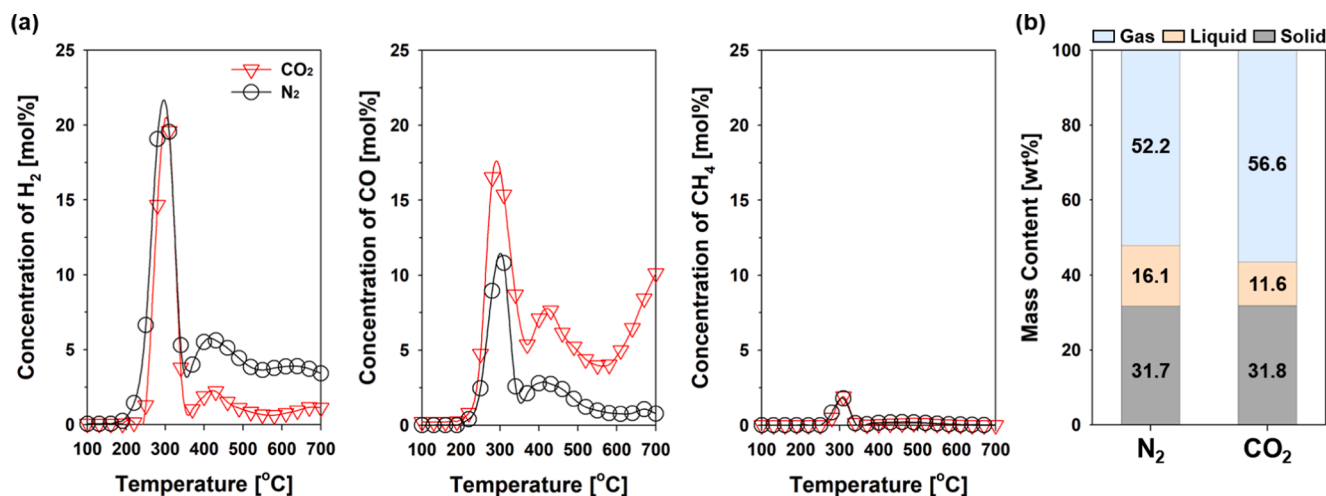


Fig. 7. (a) Concentration curves of major pyrogenic gases, and (b) mass balances of three pyrogenic products (gas, liquid, and solid) from the catalytic pyrolysis of RTR over Ni/Al<sub>2</sub>O<sub>3</sub> under N<sub>2</sub> and CO<sub>2</sub> atmospheres.

(500–700 °C) occurring the functionality of CO<sub>2</sub> for producing CO during the single-stage pyrolysis for RTR. This configuration expected enhancements for the thermal cracking of VMs (derived from the single-stage one) and the subsequent GPR with CO<sub>2</sub>. Major gaseous products (H<sub>2</sub>, CO, and CH<sub>4</sub>) were measured using an online gas monitoring unit. Their concentration curves were plotted as a function of reaction temperature (Fig. 6(a)). Mass balances of three pyrogenic products from the multi-stage pyrolysis of RTR under both atmospheres were presented in Fig. 6(b).

Overall evolution patterns for H<sub>2</sub>, CO, and CH<sub>4</sub> from the multi-stage pyrolysis were similar to those from the single-stage pyrolysis. However, applying an additional temperature (600 °C) to VMs notably increased the gas concentration levels at ≤ 500 °C. This is likely attributed to the enhanced thermal cracking of VMs. Nevertheless, the concentration curves from the N<sub>2</sub> and CO<sub>2</sub> atmospheres at ≤ 500 °C were nearly identical. This observation signifies that the GPRs of CO<sub>2</sub> with VMs were not activated at 600 °C. These results were well reflected in the mass balance results in Fig. 6(b). The biocrude yield under the N<sub>2</sub> condition was reduced from 45.7 wt% to 28.1 wt% (Fig. 6(b)). However, the conversion efficiency of biocrude by CO<sub>2</sub> for gaseous products from multi-stage pyrolysis was nearly comparable to that from single-stage pyrolysis. To address the slow reaction kinetics of CO<sub>2</sub>, a nickel-based

catalyst (Ni/Al<sub>2</sub>O<sub>3</sub>) was adopted for expediting the thermal cracking of VMs and simultaneous GPR. Nickel was chosen due to its capability to promote the thermolysis of chemical bonds in carbonaceous materials [78]. The concentration curves for H<sub>2</sub>, CO, and CH<sub>4</sub> are shown in Fig. 7 (a). Mass balances of three pyrogenic products derived from the catalytic pyrolysis of RTR under both atmospheric conditions are shown in Fig. 7 (b).

Under N<sub>2</sub> conditions, the introduction of Ni catalyst to the pyrolysis system resulted in a significant increase in syngas (Fig. 7(a)). The concentrations of H<sub>2</sub> and CO at 310 °C were, respectively, 19.54 mol% and 10.82 mol%, which increased 7.02 and 1.14-fold each compared to the results from multi-stage pyrolysis of RTR. The particular increase in the H<sub>2</sub> yield is attributed to the capability of Ni metal to preferentially activate C–H bonds rather than C–O/C–C bonds [79], promoting dehydrogenation at 600 °C [80]. The synergistic use of Ni catalyst and CO<sub>2</sub> altered the evolution trends for H<sub>2</sub> and CO. A significant enhancement of CO was developed across the entire temperature regime, while the suppression of H<sub>2</sub> was observed only at ≥ 500 °C. As mentioned in Section 3.2, the increased formation of CO and simultaneous decrease of H<sub>2</sub> at ≥ 500 °C is closely linked to the rWGS reaction. The production of CO exceeding the H<sub>2</sub> consumption cannot be explained except for the occurrence of the GPR. The enhanced yield of CO and the similar yield of



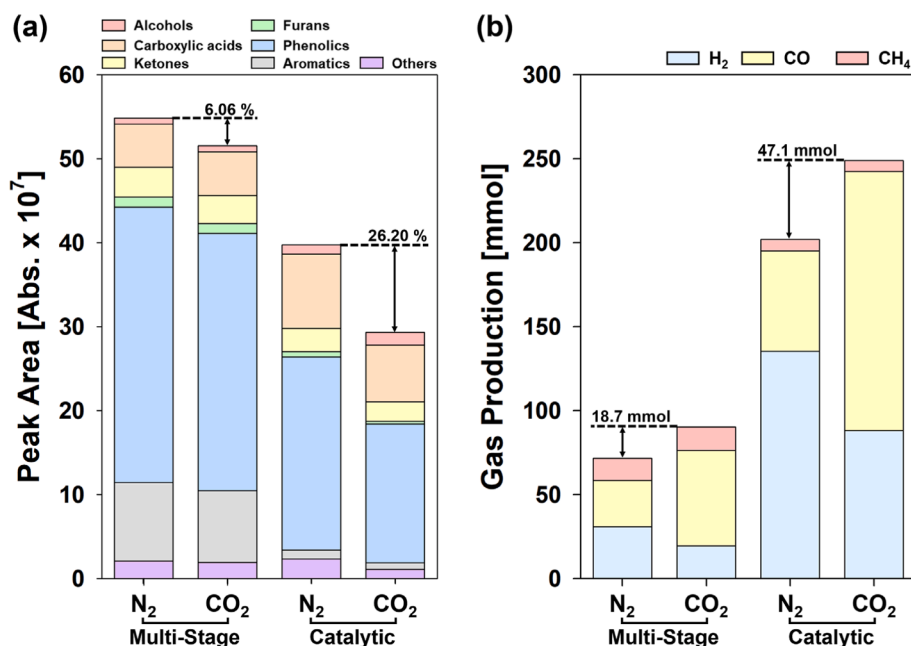


Fig. 8. (a) Total peak areas of the major chemical groups in the biocrudes, and (b) cumulative production of major pyrogenic gases derived from multi-stage and catalytic pyrolysis of RTR under N<sub>2</sub> and CO<sub>2</sub> conditions.

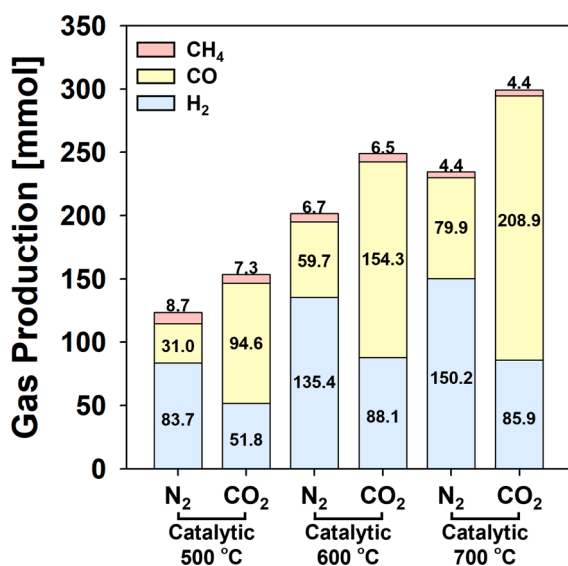


Fig. 9. Cumulative production of major pyrogenic gases (H<sub>2</sub>, CO, and CH<sub>4</sub>) obtained from catalytic pyrolysis for RTR at the isothermal temperatures of 500, 600, and 700 °C.

H<sub>2</sub> under both atmospheres were observed at ≤ 500 °C. The evolution trends obviously support the functional effectiveness of CO<sub>2</sub> in the GPR for the additional production of CO without the contribution of H<sub>2</sub>. Therefore, the exploitation of Ni catalyst in the CO<sub>2</sub>-assisted pyrolysis process facilitated the conversion of VMs (derived from RTR) into CO. The effectiveness of CO<sub>2</sub> was shown from the mass balance results that exhibited a reduction of biocrude and a simultaneous increase of gaseous products (Fig. 7(b)).

To provide a further explanation of the functional effectiveness of CO<sub>2</sub>, the compositional matrices of biocrudes from the multi-stage/catalytic pyrolysis under both atmospheres were compared. The chemical constituents in biocrudes were qualified using a GC/MS unit (data not shown). The identified chemical constituents were grouped under diverse functional linkages and their total peak areas are presented in

Fig. 8(a). In comparison to single-stage pyrolysis (Fig. 5(b)), the formation of major chemical groups notably decreased by exerting additional heat. The thermo-chemical reduction of the major chemical groups was effectively enhanced in the presence of a catalyst, and the use of CO<sub>2</sub> in the multi-stage pyrolysis of RTR showed a slight decrease in chemical constituents (6.06 area%). Simultaneous utilization of CO<sub>2</sub> and Ni catalyst resulted in 26.20 area% reduction in the total peak areas of major chemical constituents. These results correspond well to the functional reactivity of CO<sub>2</sub> as a reactive gas for partially oxidizing VMs liberated from RTR and simultaneously producing CO.

To effectively compare the efficacy of CO<sub>2</sub> on enhancing syngas production, the cumulative productions of H<sub>2</sub>, CO, and CH<sub>4</sub> from multi-step and catalytic pyrolysis were presented in Fig. 8(b). CO<sub>2</sub> in the pyrolysis process facilitated transition of VMs into syngas, particularly CO. The CO production induced by CO<sub>2</sub> was significantly improved by adding additional heat and catalyst. To help understanding the mechanistic functions of CO<sub>2</sub>, estimated reaction routes were visualized in Fig. S2. The functionality of CO<sub>2</sub> led to two main contributions for partial oxidation of VMs and subsequent enhancement of CO. Therefore, the proposed pyrolysis system offers a new opportunity to realize maximized carbon utilization.

For the in-depth investigation on the CO<sub>2</sub> functionality, catalytic pyrolysis for RTR was performed in the isothermal temperatures of 500, 600, and 700 °C. The resulting cumulative productions of H<sub>2</sub>, CO, and CH<sub>4</sub> are presented in Fig. 9. The syngas productivity proportionally increased to the isothermal temperatures under N<sub>2</sub> and CO<sub>2</sub> conditions. Notably, the enhancement of syngas induced by CO<sub>2</sub> was intensified when the reaction temperature increases. The highest syngas production was obtained at 700 °C during the CO<sub>2</sub>-assisted catalytic pyrolysis. Specifically, CO<sub>2</sub> facilitated the transition of the volatiles into syngas, particularly CO. This finding represents a compelling evidence that CO<sub>2</sub> plays a crucial role in reacting with the volatile pyrogenic products stemming from RTR. To evaluate the syngas productivity from CO<sub>2</sub>-assisted catalytic pyrolysis, this study summarized the results from other thermochemical processes (Table 3). The performances for syngas production exhibited comparable results. Interestingly, integrating CO<sub>2</sub> to the catalytic pyrolysis was effective in controlling the H<sub>2</sub> to CO ratio associated with CO increase induced by CO<sub>2</sub>. The CO production well proves that CO<sub>2</sub> served a reactive agent for enhanced carbon utilization

**Table 3**

Comparison of syngas productivity obtained from catalytic pyrolysis employed in this study and other thermochemical processes.

| No. | Feedstock                                      | Process                                       | Detailed conditions                                                                                                                   | H <sub>2</sub> and CO productivity, [mmol g <sup>-1</sup> ] | Ref.       |
|-----|------------------------------------------------|-----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|------------|
| 1   | Microalgae<br>( <i>Chlorella pyrenoidosa</i> ) | Steam gasification                            | (1) Batch reactor(2) 430 °C for 1 h(3) Sample: Catalyst = 1:2(4) 20 wt% Ni/HZMS-5                                                     | (1) H <sub>2</sub> : 5.61(2) CO: 0.63                       | [82]       |
| 2   | Almond shell                                   | Steam gasification                            | (1) Batch reactor(2) 460 °C for 0.25 h                                                                                                | (1) H <sub>2</sub> : 7.5(2) CO: 2.77                        | [83]       |
| 3   | Rice straw                                     | Chemical looping gasification                 | (1) Fixed bed reactor(2) 800 °C(3) Sample: oxygen carrier = 5.021:1(4) oxygen carrier: Ca <sub>2</sub> Fe <sub>2</sub> O <sub>5</sub> | (1) H <sub>2</sub> : 23.07(2) CO: 4.19                      | [84]       |
| 4   | Woody sawdust                                  | Pyrolysis and in-line steam reforming         | (1) Fixed bed reactor(2) 800 °C(3) Sample: catalyst = 1:1(4) 10 wt% NiAl <sub>2</sub> O <sub>4</sub> /Al <sub>2</sub> O <sub>3</sub>  | (1) H <sub>2</sub> : 4.91(2) CO: 8.79                       | [85]       |
| 5   | Wood sawdust                                   | Pyrolysis and in-line steam reforming         | (1) Fixed bed reactor(2) 800 °C(3) Sample: catalyst = 2:1(4) Ni/CaAlO <sub>x</sub>                                                    | (1) H <sub>2</sub> : 15.37(2) CO: 12.30                     | [86]       |
| 6   | Rubber tree residue                            | Catalytic pyrolysis                           | (1) Fixed bed reactor(2) 600 °C(3) Sample: catalyst = 1:1(4) 5 wt% Ni/Al <sub>2</sub> O <sub>3</sub>                                  | (1) H <sub>2</sub> : 13.5(2) CO: 5.97                       | This study |
| 7   | Rubber tree residue                            | CO <sub>2</sub> -assisted catalytic pyrolysis | (1) Fixed bed reactor(2) 600 °C(3) Sample: catalyst = 1:1(4) 5 wt% Ni/Al <sub>2</sub> O <sub>3</sub>                                  | (1) H <sub>2</sub> : 8.8(2) CO: 15.4                        | This study |

in the pyrolysis system.

This study provides an innovative approach to mitigate greenhouse gas emissions because CO<sub>2</sub> generated from industrial plants can be directly utilized in the pyrolysis system. Given that fossil fuel combustion typically evolves flue gas with 6–15 vol% CO<sub>2</sub> [81], examining the efficacy of CO<sub>2</sub> at these concentration levels offers opportunities to be further considered for practical applications of the proposed system. Nevertheless, all the experimental findings in this study were obtained under a fixed concentration of CO<sub>2</sub> (80 vol% CO<sub>2</sub>). It is crucial to explore the functionality of CO<sub>2</sub> across variable concentrations. Therefore, following studies should be tested in the near futures, to derive the optimized concentrations of CO<sub>2</sub> exhibiting its functionality in the

**Table 4**

Estimated revenue using H<sub>2</sub> produced from the integrated processes of catalytic pyrolysis and subsequent water–gas shift (WGS) reaction.

| Content                                                                    | Unit                                              | Value      | Ref.       |
|----------------------------------------------------------------------------|---------------------------------------------------|------------|------------|
| Area for global rubber plantation                                          | ha                                                | 7,728,000  | [90]       |
| Commercial lifespan of rubber tree                                         | years                                             | 30         | [41]       |
| Annual area for rubber tree harvesting                                     | ha year <sup>-1</sup>                             | 257,600    | –          |
| Estimated generation of RTR per unit area                                  | ton ha <sup>-1</sup>                              | 289.35     | [91]       |
| Annual generation of RTR                                                   | ton year <sup>-1</sup>                            | 74,536,560 | –          |
| Market price of hydrogen                                                   | US\$ kg <sup>-1</sup>                             | 2.05       | [92]       |
| H <sub>2</sub> production from conventional catalysis and WGS              | kg H <sub>2</sub> kg <sub>RTR</sub> <sup>-1</sup> | 0.0390     | This study |
| H <sub>2</sub> production from CO <sub>2</sub> -assisted catalysis and WGS | kg H <sub>2</sub> kg <sub>RTR</sub> <sup>-1</sup> | 0.0485     | This study |
| Revenue from conventional catalysis and WGS                                | Billion US\$                                      | 5.96       | This study |
| Revenue from CO <sub>2</sub> -assisted catalysis and WGS                   | Billion US\$                                      | 7.41       | This study |

proposed pyrolysis platform.

As a strategic means of emphasizing carbon utilization improved by CO<sub>2</sub>, the developed pyrolysis system in this study was integrated with well-established WGS reaction, stoichiometrically transforming CO into H<sub>2</sub> [87]. The comprehensive conversion of CO into H<sub>2</sub> offers an opportunity to notably improve economical benefits considering more expensive market price of H<sub>2</sub> [88,89]. Utilizing CO<sub>2</sub> in the integrated processes with WGS increased by 24 % in H<sub>2</sub> production (from 19.51 to 24.24 mmol kg<sub>RTR</sub><sup>-1</sup>). The potential revenue from the CO<sub>2</sub>-assisted integrated process showed a 1.24-fold increase, equivalent to US\$7.41 billion (based on Table 4). Thus, incorporating CO<sub>2</sub>-assisted catalytic pyrolysis and WGS reaction offers significant economic benefits for their practical implementation.

#### 4. Conclusions

This study focused on the valorisation of rubber RTR from large-scale rubber plantations, with the aim to enhance energy recovery. The key innovation was the introduction of CO<sub>2</sub> as a reactive agent in the pyrolysis process, with a specific emphasis on understanding the functional effectiveness of CO<sub>2</sub>. The use of CO<sub>2</sub> in the pyrolysis of RTR resulted in increased CO production at temperatures ≥500 °C, due likely to the GPR between CO<sub>2</sub> and VMs thermally liberated from RTR. Simultaneously, a reduction in biocrude was observed in the presence of CO<sub>2</sub>, indicating a potential for utilising more carbon in a pyrolysis platform. To activate the functional reactivity of CO<sub>2</sub> at ≤500 °C, the pyrolysis configurations were intentionally modified by adding thermal energy (600 °C) and a Ni/Al<sub>2</sub>O<sub>3</sub> catalyst. The use of Ni/Al<sub>2</sub>O<sub>3</sub> in the presence of CO<sub>2</sub> enabled enhanced syngas production across the entire test temperatures. The increase in CO at ≤500 °C demonstrated the occurrence of homogeneous GPR between CO<sub>2</sub> and VMs. Catalytic pyrolysis under CO<sub>2</sub> conditions resulted in increased yields of H<sub>2</sub> and CO compared to N<sub>2</sub> conditions. To emphasise the technical advantage, CO<sub>2</sub>-assisted catalytic pyrolysis was integrated/simulated with the WGS reaction to maximize energy recovery. In the thermo-chemical valorisation of RTR, the estimated revenue from H<sub>2</sub> produced in the CO<sub>2</sub>-assisted integrated process was US \$7.41 billion, representing a 1.24-fold increase compared to the conventional integrated process.

#### CRediT authorship contribution statement

**Sangyoon Lee:** Writing – original draft, Investigation, Data curation. **Taewoo Lee:** Writing – original draft, Formal analysis, Data curation. **Hoyeon Cha:** Investigation, Formal analysis, Data curation. **Sungyup Jung:** Investigation, Formal analysis, Data curation. **Yiu Fai Tsang:** Investigation, Data curation. **Jaewon Lee:** Writing – original draft, Formal analysis, Data curation. **Eilhann E. Kwon:** Writing – review & editing, Supervision, Conceptualization.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

Data will be made available on request.

## Acknowledgements

This work was supported by the National Research Foundation of Korea (NRF) grant funded by the Korean Government (MSIT) (Grant No. NRF-2023R1A2C3003011). This work was also supported by Basic Science Research Program through the National Research Foundation of Korea (NRF) funded by the Ministry of Education (Grant No. RS-2023-00274658).

## Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.enconman.2024.118554>.

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